

A Study Companion to Threshold Dynamics

From the heat equation to volume-constrained auction dynamics,
and how each idea appears in `src/partition/mbo_auction.py`

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Abstract

A learning document, written while reading the papers behind approach B (`docs/explanations/reading-path-for-a-collaborator.md`). It is aimed at a reader trained in physics and applied mathematics who wants the *intuition* behind each method well enough to read the implementation, not the proofs. Each chapter builds one idea from what a physicist already knows, works the key derivation by hand, checks it with a small numerical experiment, and ends by mapping the idea onto our code. The rigorous account of the method as implemented is `docs/math/10-mbo-auction-dynamics/`; this document is the road to it.

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How to use this document

Order. The chapters follow the reading path: the unconstrained scheme (threshold dynamics, Chapter 1), then the constraint that keeps cells at equal area, then how the diffusion step is done on a triangulated surface, then the assignment machinery. Each chapter says which pages of which paper it prepares you for.

Ch.	Topic	Papers it prepares
1	Diffusion-generated motion: the MBO scheme	MBO 1992, MBO 1994, Esedoğlu–Otto 2015 §1–4
2	Constraints and prices: Lagrange multipliers, the shifted threshold	Ruuth–Wetton 2003
3	Auction dynamics: the equal-area step as an assignment problem	Jacobs–Merkurjev–Esedoğlu 2018, Bertsekas 1988
4	Diffusion on a triangulated surface, and other kernels	Dziuk–Elliott 2013, Garcia-Cardona et al. 2014, Esedoğlu–Jacobs 2017
5	The time step on a discrete domain	van Gennip et al. 2014

Only Chapter 1 is written; the rest is the plan, and will change as reading suggests.

Boxes. *Key idea* boxes state the one thing to remember. *Physics corner* boxes translate into language you already have; skip them if the translation is obvious. *In our code* boxes say where the idea lives in the repository. *Where the numbers come from* boxes give the script that regenerates every computed number and figure in the chapter.

Reproducibility. The repository rule applies here too: any number produced by running code comes from a committed script under `docs/study/scripts/` and its committed output under `docs/study/data/`. Rebuild everything with `make -C docs/study figures` and then `make -C docs/study`.

Conventions. The heat equation is $\partial_t u = \Delta u$ (diffusion constant $D = 1$; the papers sometimes keep D). Its kernel at time t in $\mathbb{R}^d$ is

$$G_t(x) = (4\pi t)^{-d/2} e^{-|x|^2/4t},$$

the density of a normal distribution with variance $2t$ in each coordinate. A set Σ has indicator χ_Σ (1 inside, 0 outside). Curvature κ is positive where a set is convex.

Chapter 1

Diffusion-generated motion: the MBO scheme

Key idea

Blur a set with the heat equation for a short time τ , then keep the points where the blurred value is at least $\frac{1}{2}$. Repeat. The boundary of the set moves inward at a speed equal to its curvature, and curvature flow is the fastest way to shorten a boundary. With N sets, blur each one and give every point to the set whose blurred value is largest. Every step lowers an energy that approximates the total boundary length, so this simple loop is a perimeter minimiser. Approach B is this loop, with one extra rule that keeps the cell areas equal.

What this chapter prepares you for. Merriman, Bence & Osher 1992 [5], §2–4 (pages 3–5 of the report): the scheme and the authors’ own argument for why it works. Merriman, Bence & Osher 1994 [6]: the multiphase version (skim, mainly the figures). Esedoğlu & Otto 2015 [2], §1–4: the energy view of §1.8–1.9. Afterwards, docs/math/10-mbo-auction-dynamics/ §4 should read as a summary of things you already know.

Where the numbers come from

Every number and figure in this chapter is produced by docs/study/scripts/ch01_mbo.py and recorded in docs/study/data/ch01.yaml (python 3.12.13, numpy 2.4.4, scipy 1.17.1, matplotlib 3.10.8; the one random draw uses seed 20261002; about 20 s to run). There are two kinds of experiment. *Gridless* ones (experiments A and D) use the exact formula for a blurred disc, so they test the continuum analysis with no discretisation error. *Grid* ones (B, C and E) run the algorithm on a periodic unit square, a flat torus, with the heat step done exactly by FFT.

1.1 The problem the scheme solves

Our goal is to cut a surface into N cells of equal area so that the total length of the cell boundaries is as small as possible. Leave the area constraint aside for this chapter and ask the simpler question: *given a set, how do you change it so that its boundary gets shorter as fast as possible?* The answer is a classical geometric flow, which §1.3 derives. MBO is a surprisingly simple way to compute it.

Two properties make the scheme attractive for our problem, and they are the reason it replaced PGD:

1. The state is always a *hard* partition: every point belongs to exactly one cell, at every step. There is no fuzzy density to be “read out” at the end, so the gap between the relaxed

solution and the partition you finally extract, which broke PGD at high N , cannot appear (docs/reference/winner_take_all_partition_gap.md).

2. One step costs one linear solve per cell, and no small parameter ε has to resolve an interface on the mesh.

1.2 Ingredient 1: the heat equation as a blur

Solve $\partial_t u = \Delta u$ in $\mathbb{R}^d$ starting from $u(\cdot, 0) = f$. The solution at time τ is a convolution:

$$u(x, \tau) = (G_\tau * f)(x) = \int_{\mathbb{R}^d} G_\tau(x - y) f(y) dy, \quad G_\tau(x) = (4\pi\tau)^{-d/2} e^{-|x|^2/4\tau}. \quad (1.1)$$

In words, the new value at x is a Gaussian-weighted average of the old values within distance about $\sqrt{\tau}$ of x . These are the facts about G_τ used in this chapter and the next ones:

1. **It is a probability density:** $G_\tau \geq 0$ and $\int G_\tau = 1$. So if $0 \leq f \leq 1$ then $0 \leq G_\tau * f \leq 1$, and $G_\tau * 1 = 1$. Blurring an indicator gives a number between 0 and 1, which you can read as “how much of the neighbourhood of x is inside the set”.
2. **Variance 2τ per coordinate.** If $X \sim G_\tau$, then $\mathbb{E}[X_i^2] = 2\tau$. The blur radius is $\sqrt{2\tau}$ in each direction. This factor of 2 is where the 2 in $R^2 = R_0^2 - 2t$ (§1.3) comes from.
3. **It is symmetric:** $G_\tau(x) = G_\tau(-x)$, so $\int f(G_\tau * g) = \int (G_\tau * f)g$.
4. **Its Fourier transform is positive:** $\widehat{G_\tau}(\xi) = e^{-\tau|\xi|^2} > 0$. Blurring damps every wavelength and never reverses one. This is the property that makes the energy of §1.8 concave.
5. **Semigroup:** $G_s * G_t = G_{s+t}$. Blurring for s and then for t is the same as blurring for $s + t$.

Physics corner

u is a temperature, f the initial temperature and τ the elapsed time (with $D = 1$). Equivalently, $G_\tau * f$ is the expected value of f seen by a random walker started at x after time τ . When $f = \chi_\Sigma$, the blurred value $(G_\tau * \chi_\Sigma)(x)$ is *the probability that a Brownian particle started at x is inside Σ at time τ* . Both the curvature result and the energy below are easiest to see in this picture.

1.3 Ingredient 2: curvature, and the flow that shortens curves

Take a closed curve in the plane, the boundary of a set Σ , with arc length s . Its curvature $\kappa(s)$ is the rate at which the tangent direction turns per unit length. A circle of radius R has $\kappa = 1/R$, and a straight line has $\kappa = 0$. We take $\kappa > 0$ where Σ is locally convex.

How length changes when the curve moves. Push every point of the curve outward along the normal by a small distance $\phi(s)$. To first order, the length changes by

$$\delta L = \int \kappa \phi ds. \quad (1.2)$$

Check this on a circle: pushing it out uniformly by ϕ turns length $2\pi R$ into $2\pi(R + \phi)$, so $\delta L = 2\pi\phi = \int (1/R) \phi ds$. In general, a curved piece of boundary is like a piece of a circle of radius $1/\kappa$; pushing it outward lengthens it in proportion to κ , while moving a straight piece changes nothing.

Steepest descent. Now let the curve move with outward normal velocity $V(s)$. Then $dL/dt = \int \kappa V ds$. Among all velocity fields of a given size $\int V^2 ds$, this rate is most negative for $V = -\kappa$:

move every point inward at a speed equal to its curvature. That is **motion by curvature** (curve-shortening flow), and along it

$$\frac{dL}{dt} = - \int \kappa^2 ds \leq 0. \quad (1.3)$$

Curvature flow is the gradient flow of length.

Two exact consequences, which the experiments will check.

- *Area falls at a constant rate.* $dA/dt = \int V ds = - \int \kappa ds = -2\pi$ for every simple closed curve, because the tangent turns through exactly 2π going once around. The rate does not depend on the shape.
- *A disc stays a disc.* With $dR/dt = -1/R$,

$$R(t)^2 = R_0^2 - 2t, \quad \text{extinct at } t^* = R_0^2/2. \quad (1.4)$$

In the plane, curve-shortening flow never pinches an embedded curve into two pieces. Every such curve becomes convex and then shrinks to a round point (Grayson's theorem [3]). Experiment C shows this. On a surface the same flow is *geodesic* curvature flow, and the formulas change only through the surface's own curvature.

Physics corner

This is the law of motion of a grain boundary or a soap film, $v = \mu\sigma\kappa$ (mobility times surface tension times curvature), which Mullins wrote down for idealised grain growth [8]. In a 2-D cellular network with 120° junctions it gives the von Neumann–Mullins law: a cell with n sides changes area at a rate proportional to $n - 6$. Cells with fewer than six sides shrink, cells with more grow, and hexagons are neutral. Keep this in mind for Chapter 3, where auction dynamics tiles a flat torus with hexagons.

1.4 The MBO algorithm

Two-phase MBO [5]

Given a set Σ^0 and a time step τ , repeat for $n = 0, 1, 2, \dots$:

1. **Diffuse:** $u = G_\tau * \chi_{\Sigma^n}$.
2. **Threshold:** $\Sigma^{n+1} = \{x : u(x) \geq \frac{1}{2}\}$.

After n steps the boundary approximates the curvature flow at time $t = n\tau$.

The diffuse step is linear and cheap (an FFT on a grid, one sparse solve on our mesh). The threshold step is pointwise. Nothing tracks the curve, so corners, merging and topology changes need no special handling.

Physics corner

Think of it as a repeated quench: let heat spread for a short time, then quench to zero temperature, so every point snaps to whichever phase is locally in the majority. The physics of the interface (its motion by curvature) emerges from the competition between the two operations. The interface itself is never represented.

1.5 Why the boundary moves by curvature

1.5.1 The argument in the 1992 paper

MBO's own argument [5, §3] is the one to read first. At a boundary point P , use polar coordinates (r, θ) centred at the centre of curvature of P , so P is at $r = \rho = 1/\kappa$. The heat equation reads

$$\partial_t \chi = \partial_{rr} \chi + \frac{1}{r} \partial_r \chi + \frac{1}{r^2} \partial_{\theta\theta} \chi.$$

Near P the set is locally a disc around that centre, so θ -derivatives vanish and

$$\partial_t \chi - \frac{1}{r} \partial_r \chi = \partial_{rr} \chi. \quad (1.5)$$

This is a one-dimensional *advection–diffusion* equation. The left side is transport toward the centre of curvature at speed $1/r = \kappa$. The right side is plain 1-D diffusion, which smears the step symmetrically about its midpoint and therefore does not move the level $\chi = \frac{1}{2}$. So in a short time τ the $\frac{1}{2}$ level is carried a distance $\kappa\tau$ toward the centre of curvature, and thresholding at $\frac{1}{2}$ reads off exactly that level. Figure 3 of the paper shows this picture: a step that advects and spreads.

1.5.2 The same result from the Gaussian directly

The polar argument is heuristic: the step is not smooth, and “locally a disc” needs care. A short computation makes it concrete, and it is the kind of estimate the later papers use. Put P at the origin with the set below a parabola, $\Sigma = \{(x, y) : y < -\frac{\kappa}{2}x^2\}$ (convex for $\kappa > 0$). Evaluate the blurred indicator at the point $(0, y_0)$. A 2-D Gaussian is a product of two 1-D Gaussians g_τ , each with variance 2τ , so with $\Phi_\tau(z) = \int_{-\infty}^z g_\tau$,

$$u(0, y_0) = \Pr[y_0 + Y < -\frac{\kappa}{2}X^2] = \int g_\tau(x) \Phi_\tau(-\frac{\kappa}{2}x^2 - y_0) dx.$$

The typical x is of size $\sqrt{\tau}$, so the argument of Φ_τ is of size $\kappa\tau + |y_0|$. That is much smaller than the blur width $\sqrt{\tau}$ when $\kappa\sqrt{\tau} \ll 1$. So we may linearise, $\Phi_\tau(z) \approx \frac{1}{2} + g_\tau(0)z$:

$$u(0, y_0) \approx \frac{1}{2} - g_\tau(0) \left(\frac{\kappa}{2} \mathbb{E}[X^2] + y_0 \right) = \frac{1}{2} - g_\tau(0) (\kappa\tau + y_0).$$

The new boundary, where $u = \frac{1}{2}$, is at $y_0 = -\kappa\tau$. In one step the boundary moves into the set by

$$\boxed{\text{displacement} = \kappa \tau (1 + O(\kappa^2 \tau))} \implies V = \kappa. \quad (1.6)$$

A flat interface ($\kappa = 0$) does not move at all: by symmetry its blurred profile crosses $\frac{1}{2}$ exactly at the old boundary.

Experiment A (gridless): the one-step displacement. For a disc of radius R , the blurred indicator at distance r from the centre is the probability that a normal vector with variance 2τ , started at distance r , lands in the disc. This is a noncentral χ^2 distribution function, exact and with no grid. Figure 1.1 (left) compares the profile with that of a flat interface; the disc's profile crosses $\frac{1}{2}$ inside the old boundary. Figure 1.1 (right) plots the exact displacement divided by $\kappa\tau$. The three radii fall on one curve because, by dimensional analysis, the ratio can depend only on $\sqrt{\tau}/R$. For $R = 0.2$ the ratio is 1.002, 1.008, 1.036 and 1.209 at $\sqrt{\tau}/R = 0.05, 0.1, 0.2$ and 0.4 : equation (1.6) is exact in the limit, and the relative correction grows like $(\sqrt{\tau}/R)^2 = \kappa^2\tau$ (it quadruples each time $\sqrt{\tau}/R$ doubles), as the $O(\kappa^2\tau)$ in (1.6) says. It has no term linear in $\sqrt{\tau}/R$ because Φ_τ has no quadratic term at its midpoint.

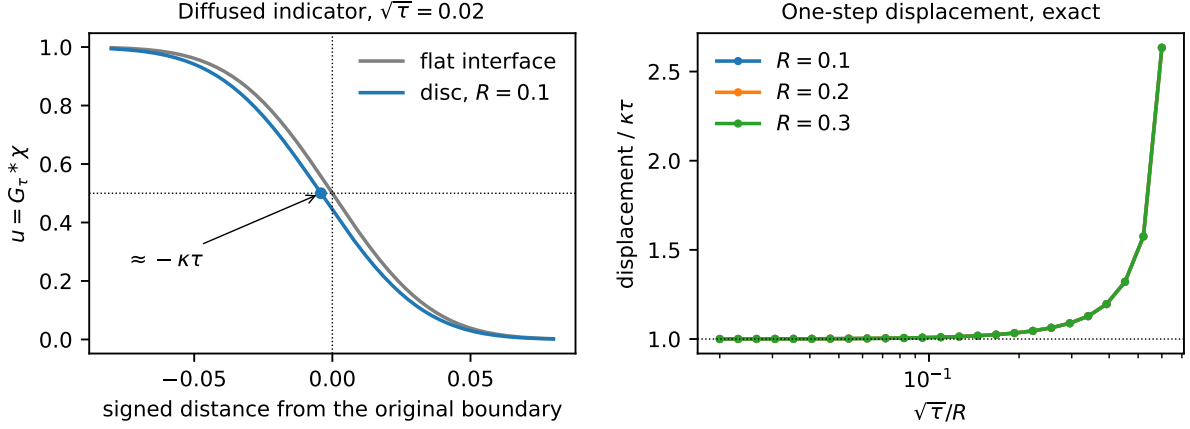


Figure 1.1: Left: blurred indicator across a flat interface and across the boundary of a disc of radius 0.1, at $\sqrt{\tau} = 0.02$. The $\frac{1}{2}$ crossing of the disc (dot) sits about $\kappa\tau$ inside the original boundary. Right: exact one-step displacement over $\kappa\tau$; the ratio tends to 1 as $\sqrt{\tau}/R \rightarrow 0$.

Experiment B (grid): a disc shrinking over many steps. On a 512×512 periodic grid ($h = 1/512 \approx 0.00195$), start from a pixelated disc with $R_0 = 0.25$ and run MBO. Equation (1.4) predicts R^2 falling linearly to zero at $t^* = 0.03125$. With $\tau = 10^{-3}$, Figure 1.3 shows the measured $R^2 = \text{area}/\pi$ on that line: the disc vanishes at $t = 0.031$, and halfway through, $R^2 = 0.0303$ against 0.0305 predicted.

Experiment C (grid): a non-convex set. Figure 1.2 runs MBO ($\tau = 5 \times 10^{-4}$) on a disc and a square joined by a bar. Corners round off immediately, concave corners fill in, the shape becomes convex as Grayson's theorem predicts, and the area falls at the shape-independent rate $2\pi \approx 6.28$. The measured mean rate is 6.47 up to $t = 0.008$ and 6.49 up to $t = 0.016$, within 3%.

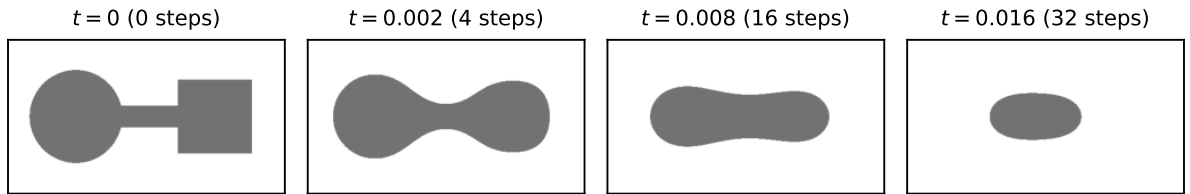


Figure 1.2: MBO on a non-convex set (512^2 grid, $\tau = 5 \times 10^{-4}$). Sharp features are smoothed first; the set becomes convex and shrinks.

1.6 The time step is a two-sided window

The only parameter is τ , and on a discrete grid it is squeezed from both sides. MBO state both constraints [5, §4]:

- **Not too large.** The analysis above needs the blur radius to be small compared with the radius of curvature, $\sqrt{\tau} \ll \rho$. Otherwise the blur “sees” far-away parts of the set and the motion is no longer local.
- **Not too small.** On a grid of spacing h , the boundary must move at least one grid cell per step, $\kappa\tau \gtrsim h$. Otherwise the thresholded set is identical to the old one and the scheme is stuck forever. This is called *pinning* or *freezing*.

With $\rho = 1/\kappa$ these combine into

$$\rho h \lesssim \tau \ll \rho^2, \quad \text{or} \quad \sqrt{\rho/h} \lesssim \sqrt{\tau}/h \ll \rho/h. \quad (1.7)$$

The window exists only if $\rho \gg h$: features must be well resolved on the grid.

Figure 1.3 shows both failures on the same disc. With $\tau = 2 \times 10^{-5}$, the displacement per step is $\kappa\tau = 0.04h$; the set never changes, and R^2 stays at 0.0625 for all 1875 steps. With $\tau = 10^{-2}$, $\sqrt{\tau}/R_0 = 0.4$; the disc shrinks too fast ($R^2 = 0.0166$ at $t = 0.02$, against 0.0225 predicted) and is gone after three steps. Only the middle value tracks the true flow.

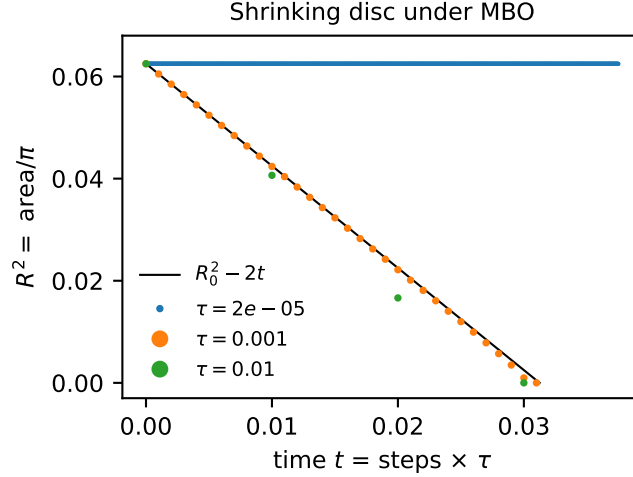


Figure 1.3: A disc of radius 0.25 under MBO on a 512^2 grid, against the exact law $R^2 = R_0^2 - 2t$. The smallest τ is pinned (no motion at all), the middle one tracks the law, and the largest overshoots.

In our code

`tau_diagnostics` in `src/partition/mbo.auction.py` enforces this window, with the radius of an equal-area cell $R_{\text{cell}} = \sqrt{\text{target area}/\pi}$ standing in for ρ and the mean edge length h_{mean} for h . The rule is $\tau = \min((c h_{\text{mean}})^2, (\rho_{\text{cap}} R_{\text{cell}})^2)$, and the reported lower edge `c_lo_hmean` $= \sqrt{R_{\text{cell}}/h_{\text{mean}}}$ is the left inequality of (1.7) written for $c = \sqrt{\tau}/h_{\text{mean}}$. Our diffusion is not a Gaussian (§1.10) and moves a boundary by $\tau\kappa/2$ rather than $\tau\kappa$, so the constants shift by a factor of order one (docs/math/10-mbo-auction-dynamics/ §5.5; the window itself is its §8). The measured window on our meshes is in docs/experiments/08-mbo-auction-dynamics/. Why pinning is unavoidable on a discrete domain is van Gennip et al. [10], Chapter 5.

1.7 Many phases: the argmax step

For a partition into N sets $\Omega_1, \dots, \Omega_N$, MBO 1994 [6] blur every indicator and give each point to the winner:

$$u_k = G_\tau * \chi_{\Omega_k}, \quad \omega^{\text{new}}(x) = \arg \max_k u_k(x). \quad (1.8)$$

Because $\sum_k \chi_{\Omega_k} = 1$ and the blur is linear, $\sum_k u_k = 1$ at every point. With $N = 2$, $u_1 > u_2$ is the same as $u_1 > \frac{1}{2}$, so (1.8) reduces to two-phase MBO. Away from junctions each interface moves by curvature as before. Where three cells meet, the scheme produces the junction configuration of the equal-tension network with no special treatment; equal tensions balance at 120° .

Experiment E (grid): sixteen cells on a flat torus, no constraint. Start from a Voronoi partition of the periodic unit square into 16 cells and run (1.8) with $\tau = 10^{-3}$ on a 256^2 grid (Figure 1.4). Small cells, and cells with few neighbours, shrink and vanish: 16 cells at step 0, 12 at step 10, 8 at step 100, and a single cell by step 400. *This is correct behaviour.* On a torus, the partition with the least total boundary is the trivial one, a single cell with no boundary at all, and unconstrained MBO is heading there. To get N cells of equal area we must *forbid* cells from shrinking. That constraint is what Ruuth–Wetton [9] add for one set and Jacobs–Merkurjev–Esedoglu [4] add for N , and it is the subject of Chapter 3.

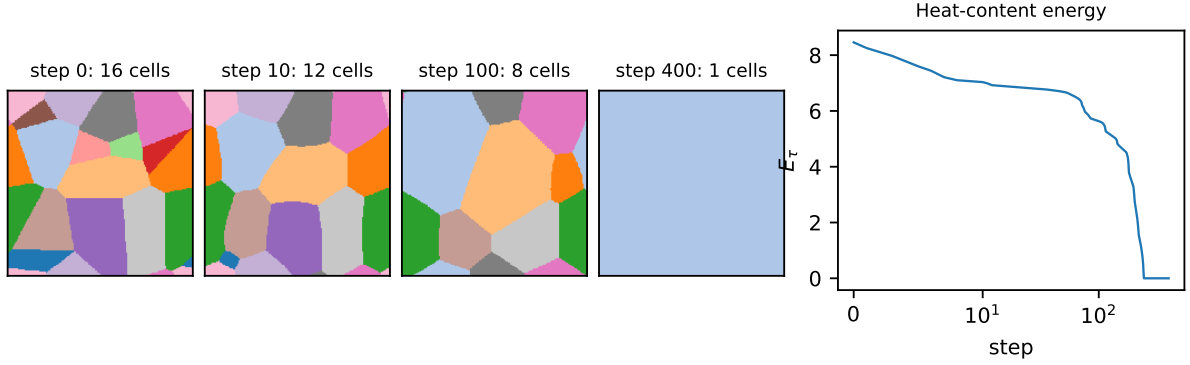


Figure 1.4: Unconstrained multiphase MBO on the flat torus (256^2 grid, 16 Voronoi cells, $\tau = 10^{-3}$). Cells vanish one by one (coarsening). Right: the heat-content energy of §1.8, which never increases.

1.8 The energy behind the scheme: heat content

So far MBO is an algorithm that happens to approximate a flow. Esedoglu and Otto [2] showed it is something stronger: each step *minimises* something. This is the view that makes the area constraint easy to add.

Two phases. Heat the set Σ to temperature 1, keep the outside at 0, and let heat flow for time τ . The heat that has left Σ is

$$E_\tau(\Sigma) = \frac{1}{\sqrt{\tau}} \int_{\Sigma} (1 - G_\tau * \chi_{\Sigma}) dA = \frac{1}{\sqrt{\tau}} \int_{\Sigma} G_\tau * \chi_{\Sigma^c} dA. \quad (1.9)$$

Heat leaks out only near the boundary, within a distance of about $\sqrt{\tau}$, so this quantity measures boundary length. Make it quantitative on a straight interface: in the direction normal to the interface, a particle that starts at depth s inside has crossed it after time τ if its 1-D displacement X (variance 2τ) exceeds s . Integrating over the starting depth,

$$\text{heat lost per unit length} = \int_0^\infty \Pr[X > s] ds = \mathbb{E}[X^+] = \frac{\sqrt{2\tau}}{\sqrt{2\pi}} = \sqrt{\frac{\tau}{\pi}},$$

using $\mathbb{E}[X^+] = \sigma/\sqrt{2\pi}$ for a centred normal with standard deviation σ . Dividing by $\sqrt{\tau}$ gives

$$E_\tau(\Sigma) \longrightarrow \frac{1}{\sqrt{\pi}} \text{Per}(\Sigma) \quad (\tau \rightarrow 0). \quad (1.10)$$

The $1/\sqrt{\tau}$ in front of (1.9) exists to make this limit finite.

Experiment D (gridless). For a disc of radius 0.2 the limit is $2\pi R/\sqrt{\pi} = 0.70898$. The exact heat content divided by that limit is 0.957, 0.990, 0.9975, 0.99937 and 0.99984 at $\sqrt{\tau}/R = 0.4, 0.2, 0.1, 0.05$ and 0.025 , converging to 1.

Many phases. The natural generalisation adds up the heat each cell loses:

$$E_\tau(\omega) = \frac{1}{\sqrt{\tau}} \sum_{k=1}^N \int \chi_{\Omega_k} (1 - G_\tau * \chi_{\Omega_k}) dA \longrightarrow \frac{1}{\sqrt{\pi}} \sum_{k=1}^N \text{Per}(\Omega_k). \quad (1.11)$$

Every interface between two cells is counted twice, once from each side. This is the same convention as our project's perimeter $\sum_k \text{Per}(\text{cell}_k)$ (see the standing rule in `CLAUDE.md`). For $N = 2$ it is twice the two-phase energy (1.9).

“ Γ -convergence” in plain words. The limit (1.10) holds for each fixed smooth set. What optimisation needs is stronger: that *minimisers* of E_τ approach minimisers of perimeter. Γ -convergence is the notion of convergence built to guarantee this. Informally it asks for two things. (i) No cheating from below: any sequence of sets approaching a shape costs, in the limit, at least that shape's perimeter. (ii) Every shape's perimeter is achieved: some sequence approaching the shape has exactly that limiting cost. Esedoglu and Otto prove this for the heat content [2, App. A], and Esedoglu–Jacobs [1] extend it to other kernels. You can treat the theorem as the licence to say that E_τ is a perimeter.

Physics corner

This is the same story as Allen–Cahn / Ginzburg–Landau. The phase-field energy $\int \varepsilon |\nabla u|^2 + \varepsilon^{-1} W(u)$, with a double-well potential W , Γ -converges to (surface tension) $\times$ (interface length) as the interface width $\varepsilon \rightarrow 0$ (Modica–Mortola [7]). That is the relaxation Bogosel–Oudet use, and so it is what our PGD Phase 1 minimises. The heat content is a *nonlocal* cousin, with interface width $\sqrt{\tau}$ in place of ε . Both methods minimise a surrogate for perimeter. The difference is the variable: PGD optimises a fuzzy density and must read a partition off at the end, while MBO only ever holds a partition.

1.9 Each MBO step is a descent step

This section is the bridge to auction dynamics, so it is worth understanding fully. It needs only two inequalities.

Relax. Let the cells be “soft”: $u = (u_1, \dots, u_N)$ with $u_k(x) \geq 0$ and $\sum_k u_k(x) = 1$ at each point (a probability vector per point). Write the energy for such u . Since $\sum_k \int u_k = |\Omega|$ (the total area),

$$E_\tau(u) = \frac{1}{\sqrt{\tau}} \left[|\Omega| - \sum_k \int u_k (G_\tau * u_k) dA \right]. \quad (1.12)$$

For hard indicators this is exactly (1.11).

It is concave. By Parseval, $\int f (G_\tau * f) = (2\pi)^{-d} \int \widehat{G_\tau}(\xi) |\hat{f}(\xi)|^2 d\xi \geq 0$, because $\widehat{G_\tau} = e^{-\tau|\xi|^2} > 0$. So each term $\int u_k G_\tau * u_k$ is a positive quadratic form in u_k , a bowl. Since it enters with a minus sign, E_τ is an upside-down bowl: *concave*. This is property 4 of §1.2 at work.

Linearise. At the current partition u , the tangent-plane approximation of E_τ is

$$L_u(u') = E_\tau(u) + \langle \nabla E_\tau(u), u' - u \rangle, \quad \nabla_{u_k} E_\tau(u) = -\frac{2}{\sqrt{\tau}} G_\tau * u_k,$$

using the symmetry of G_τ (property 3). Apart from constants, minimising L_u means maximising $\sum_k \int u'_k (G_\tau * u_k)$ over probability vectors u' . This decouples point by point. At each x , put all the weight on the largest $(G_\tau * u_k)(x)$. **That is exactly the argmax step (1.8).** So an MBO step is “minimise the tangent plane of the energy”.

Descend. For a concave function, the tangent plane lies *above* the graph everywhere. So, with u' the MBO update,

$$E_\tau(u') \leq L_u(u') \leq L_u(u) = E_\tau(u). \quad (1.13)$$

The first inequality is concavity. The second holds because u' minimises L_u and u is one of the candidates. Hence **the energy never increases**, for any τ , with no step-size control. This is Esedoğlu–Otto’s Lemma 5.2 in our notation [2]. On the grid of Experiment E the FFT blur also has a positive multiplier, so the argument applies verbatim. The measured energy falls from 8.46 to 0, and no step increases it (Figure 1.4, right).

Key idea: why this is the door to auction dynamics

The second inequality in (1.13) used only that the current partition u is *one of the allowed candidates*. So add any constraint the current partition already satisfies, such as “every cell has area A ”, and minimise L_u over the constrained candidates. The energy still cannot increase, provided the constrained problem is solved *exactly*. The price is that the problem no longer decouples point by point: the cells compete for a fixed budget of area. Chapter 3 shows that it becomes a classical *assignment problem*. Its solution is a shifted argmax $\arg \max_k [u_k(x) + \psi_k]$, where each ψ_k is a Lagrange multiplier, a “price” for cell k .

Physics corner

The ψ_k are chemical potentials. Fixing each cell’s area is like fixing each species’ particle number. Lagrange multipliers turn the constrained (canonical) problem into an unconstrained (grand-canonical) one, and the multiplier is set so that the average particle number comes out right. A cell that is too small raises its ψ_k and attracts points; a cell that is too large lowers it.

1.10 From the flat torus to our code

In this chapter	In <code>src/partition/mbo_auction.py</code>
Indicators χ_{Ω_k}	<code>sparse_one_hot(labels, n_cells)</code> ; the state is a vertex labelling
Blur $G_\tau * \chi_k$ (exact heat flow)	<code>diffuse_indicators</code> : solve $(M + \tau K) y_k = M \chi_k$ for each k , one backward-Euler step of $M \dot{y} + K y = 0$, with M, K the P1 mass and stiffness matrices
Argmax (1.8)	<code>balanced_assign</code> : the <i>shifted</i> argmax $\arg \max_k [y_{ik} + \psi_k]$ with equal-area prices (Chapter 3)
Heat content (1.12)	<code>lyapunov_energy</code> : $(S - \sum_k \chi_k^\top D y_k) / \sqrt{\tau}$, the same formula with integrals replaced by lumped-mass sums
Descent inequality (1.13)	<code>descent_slack</code> : what survives of it when the assignment is solved inexactly (docs/math/10-mbo-auction-dynamics/§7)
Time-step window (1.7)	<code>tau_diagnostics</code>

Why our blur is not a Gaussian, and why that is fine. On a triangulated surface there is no FFT. The code takes one implicit-Euler step of the heat equation, $(I - \tau\Delta)^{-1}$ in continuum notation, rather than the exact flow $e^{\tau\Delta}$. In Fourier language the multiplier is $1/(1 + \tau|\xi|^2)$ instead of $e^{-\tau|\xi|^2}$. The two agree for long wavelengths, and both are *positive*, so the concavity argument of §1.9 survives. There is also a neat identity,

$$\frac{1}{1 + \tau|\xi|^2} = \int_0^\infty e^{-s} e^{-s\tau|\xi|^2} ds,$$

which says that one implicit step is an *average of exact heat flows* over random times $s\tau$ with s exponentially distributed. Averaging (1.6) and (1.10) over those times is how docs/math/10-mbo-auction-dynamics/ §5.2–5.5 obtains our kernel’s constants (per-step displacement $\tau\kappa/2$, interface constant $\frac{1}{2}$ in place of $1/\sqrt{\pi}$). This is also why Esedoğlu–Jacobs [1], on kernels other than the Gaussian, matters to us (Chapter 4).

1.11 Exercises

Short, and meant to be done with pencil or with the script.

1. From $dR/dt = -1/R$, derive (1.4). A square of side a under curvature flow: when does it vanish? (Hint: use $dA/dt = -2\pi$, not the shape.)
2. Show $\mathbb{E}[X^+] = \sigma/\sqrt{2\pi}$ for $X \sim N(0, \sigma^2)$, and recover the constant $1/\sqrt{\pi}$ in (1.10).
3. In one dimension a “set” is an interval of length ℓ . Show that two-phase MBO with $\sqrt{\tau} \ll \ell$ moves its endpoints by an amount that is exponentially small in ℓ^2/τ . (There is no curvature in 1-D; only the far endpoint is felt.) What happens when $\ell \lesssim \sqrt{\tau}$?
4. Threshold at 0.4 instead of $\frac{1}{2}$. How does a flat interface move now, and how far per step? You have just added a pressure term. Ruuth–Wetton [9] choose the threshold *at every step* so that the area is preserved. That is the one-cell version of the prices ψ_k .
5. Check that for $N = 2$, (1.11) equals twice (1.9).
6. With the script, lower τ in experiment B step by step and locate the pinning threshold on the 512^2 grid. Compare it with $\kappa\tau \approx h$.

1.12 Summary

- Curvature flow is the steepest descent of boundary length (1.3).
- MBO computes it by *blur, then threshold at $\frac{1}{2}$* . Each step moves the boundary by $\kappa\tau$ (1.6), and τ must sit in a two-sided window (1.7).
- With N cells the threshold becomes an argmax (1.8). Left alone, cells vanish, because fewer cells means less boundary.
- MBO minimises the tangent plane of a concave heat-content energy that converges to the total perimeter, so it never increases that energy (1.13). The argument tolerates any constraint the current partition already satisfies, which is how the equal-area constraint enters in Chapter 3.

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